

Day-9

DATE:18-08-25

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Hello all, today's task below:

- **Create Fast API, to store, retrieve all and retrieve specific song by Id.**
 - **SQLite can be used as data store**
 - **table can contain**
 - **song name, playing time in seconds, album (or movie) along with Id**

[Day10task.py](#):

```
from sqlalchemy import create_engine

from sqlalchemy.dialects.sqlite import *

SQLALCHEMY_DATABASE_URL = "sqlite:///./songs.db"

from typing import List

from fastapi.templating import Jinja2Templates

from fastapi.responses import HTMLResponse

engine = create_engine(SQLALCHEMY_DATABASE_URL
,connect_args={"check_same_thread":False})

from sqlalchemy.orm import sessionmaker,Session ,declarative_base

session = sessionmaker(autocommit=False,autoflush=False,bind=engine)

Base = declarative_base()

from sqlalchemy import Column,Integer,String

class song(Base):

    __tablename__ ='songs'

    Id = Column(Integer,primary_key=True,index=True)

    Song_Name= Column(String(50),nullable=False)
```

```
Duration = Column(Integer,nullable=False)

Base.metadata.create_all(bind=engine)

from fastapi import FastAPI, Request, Depends

import uvicorn

app = FastAPI()

def get_db():

    db=session()

    try:

        yield db

    finally:

        db.close()

from pydantic import BaseModel

class SongSchema(BaseModel):

    Id:int

    Song_Name:str

    Duration:int

class Config:

    orm_mode=True

@app.post('/add_song')

def add_song(s:SongSchema,db:Session=Depends(get_db)):

    # db=get_db()

    st = song(Song_Name=s.Song_Name, Duration=s.Duration)
```

```

    db.add(st)

    db.commit()

    db.refresh(st)

    return st

templates = Jinja2Templates(directory="templates")

@app.get("/list_html", response_class=HTMLResponse)
def get_all_songs_html(request: Request, db: Session = Depends(get_db)):

    recs = db.query(song).all()

    return templates.TemplateResponse("songlist.html", {"request": request, "songs":
recs})

class song_dto(BaseModel):

    Name:str

    class Config:

        orm_mode=True

@app.get("/getBy/{my_id}",response_model=song_dto)
async def getBy(my_id:int,db:Session=Depends(get_db)):

    st=db.query(song).filter_by(Id=my_id).first()

    st_dto = song_dto(Name=st.Song_Name)

    return st_dto

@app.middleware("http")
async def addmiddleware(request:Request,call_next):

    print("Middleware has intercepted the calls!")

    response = await call_next(request)

    return response

```

```
if __name__=="__main__":  
    uvicorn.run("Day10task:app",host="127.0.0.1",port=8000,reload=True)
```

Songlist.html:

```
<!DOCTYPE html>  
  
<html>  
  
<head>  
  
    <title>Songs List</title>  
  
    <style>  
        table {  
            border-collapse: collapse;  
            width: 60%;  
            margin: 20px auto;  
        }  
        th, td {  
            border: 1px solid black;  
            padding: 8px;  
            text-align: center;  
        }  
        th {  
            background-color: #f2f2f2;  
        }  
    </style>  
</head>  
  
<tbody>  
    <tr>  
        <td>Song Name</td>  
        <td>Artist</td>  
        <td>Album</td>  
    </tr>  
</tbody>  
</table>  
</tbody>  
</html>
```

```
</style>

</head>

<body>

  <h2 style="text-align:center;">Songs List</h2>

  <table>

    <tr>

      <th>ID</th>

      <th>Song Name</th>

      <th>Duration</th>

    </tr>

    {% for s in songs %}

      <tr>

        <td>{{ s.Id }}</td>

        <td>{{ s.Song_Name }}</td>

        <td>{{ s.Duration }}</td>

      </tr>

    {% endfor %}

  </table>

</body>

</html>
```

ADD Song

Edit Value | Schema

```
{
  "Id": 0,
  "Song_Name": "visiri",
  "Duration": 240
}
```

Execute

Database Structure | Browse Data | Edit Pragmas | Execute SQL

Table: songs

	<u>Id</u>	Song_Name	Duration
	Filter	Filter	Filter
1	1	anbil avan	180
2	2	ennai konjam ...	240
3	3	Annal Meelae	180
4	4	Vaarayo Vaarayo	300
5	5	visiri	240

Display songs:

GET

/list_html

Get All Songs Html

Parameters

No parameters

Execute

Responses

Curl

```
curl -X 'GET' \
  'http://127.0.0.1:8000/list_html' \
  -H 'accept: text/html'
```

Request URL

```
http://127.0.0.1:8000/list_html
```

Server response

Code

Details

Songs List

ID	Song Name	Duration
1	anbil avan	180
2	ennai konjam maatri	240
3	Annal Meelae	180
4	Vaarayo Vaarayo	300
5	visiri	240

GetbyId:

Parameters

Name	Description
my_id * required integer (path)	3

Execute

Responses

Curl

```
curl -X 'GET' \
'http://127.0.0.1:8000/getBy/3' \
-H 'accept: application/json'
```

Request URL

```
http://127.0.0.1:8000/getBy/3
```

Server response

Code	Details
200	Response body <pre>{ "Name": "Annal Meelae" }</pre> Response headers <pre>content-length: 23 content-type: application/json date: Mon, 18 Aug 2025 12:06:29 GMT server: uvicorn</pre>

Responses

